

10. **WAAM is commonly used in which application?**
 - A) Printing large metal structures
 - B) Manufacturing ceramic parts
 - C) Printing electronic circuits
 - D) Producing plastic prototypes
11. **Why is preheating used in laser powder bed fusion for some metals?**
 - A) To increase cooling rate
 - B) To reduce thermal stresses and cracking
 - C) To speed up printing
 - D) To enhance powder flowability
12. **What does omnidirectionality mean in WAAM?**
 - A) Printing in all directions
 - B) Using multiple lasers
 - C) Rotating the wire feeder
 - D) Changing the printing material dynamically
13. **What is toolpath planning?**
 - A) Defining the movement of the print head or laser
 - B) Selecting the material for printing
 - C) Generating the STL file
 - D) Cleaning the printed part
14. **Which AM process typically requiring the least support structures?**
 - A) Stereolithography (SLA)
 - B) Fused Deposition Modelling (FDM)
 - C) Selective Laser Sintering (SLS)
 - D) Binder Jetting
15. **What is a significant geometric limitation of FDM?**
 - A) Limited colour options
 - B) Need for support structures
 - C) Large build volume
 - D) High print speed
16. **Which industry benefits most from laser beam powder bed AM?**
 - A) Automotive
 - B) Aerospace
 - C) Textile
 - D) Food
17. **What is the primary purpose of de-cubing in laminated object manufacturing (LOM)**
 - A) To remove excess resin
 - B) To separate the part from unused material
 - C) To fuse layers together
 - D) To cure the part under UV light
18. **Which post-processing technique is used to improve the surface finish of metal AM parts?**
 - A) Shot peening
 - B) Heat treatment
 - C) Sandblasting
 - D) All of the above
19. **What is the primary cause of residual stresses in WAAM?**
 - A) Rapid solidification
 - B) Slow cooling rate
 - C) Inert gas flow
 - D) Powder particle size
20. **What is the primary cause of porosity in WAAM?**
 - A) Lack of fusion
 - B) Excessive heating
 - C) Low scanning speed
 - D) High gas pressure

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is digital manufacturing? Discuss the generative steps in the additive manufacturing with the help of a flowchart	8	L*1	1	1,3
	b) Explain laser bed powder additive manufacturing with neat sketch	8	L1	2	1,3

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|-------|---|---|----|---|-----|
| 2. a) | Write a note on the following | | | | |
| | • Support generation by facet | 8 | L1 | 1 | 1,3 |
| | • Support generation by contours | | | | |
| b) | Explain the binder jetting process briefing its working principle, apparatus, process and application | 8 | L1 | 2 | 1,3 |
| 3. a) | What is STL? Explain the data structure of STL files in ASCII and Binary formats | 8 | L1 | 1 | 1,3 |
| b) | Explain directed energy process briefing its working principle, apparatus, process and application | 8 | L1 | 2 | 1,3 |

Unit- II

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|-------|---|---|----|---|-----|
| 4. a) | Explain following approaches with respect to sheet metal additive manufacturing | | | | |
| | i. Form and then Bond | | | | |
| | ii. Bond and then Form | 8 | L1 | 3 | 1,3 |
| b) | Discuss Cost Minimization Model with respect to AM | 8 | L1 | 4 | 1,3 |
| 5. a) | What is omnidirectionality? Suggest any three ways to achieve omnidirectionality in additive manufacturing | 8 | L1 | 3 | 1,3 |
| b) | Discuss the following with respect to powder bed additive manufacturing material capabilities | | | | |
| | • Direct route | | | | |
| | • Indirect route | 8 | L1 | 4 | 1,3 |
| 6. a) | Discuss briefly about the geometric and material capability of additive manufacturing | 8 | L1 | 3 | 1,3 |
| b) | Briefly explain the effect of variation in solidification rate on the microstructure, hardness and tensile properties in AM with respect to WAAM. | 8 | L1 | 4 | 1,3 |

Unit-III

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|-------|--|---|----|---|-----|
| 7. a) | How are support structures removed in the following process? | | | | |
| | i. Rapid freezing | | | | |
| | ii. Stereolithography (SLA) | | | | |
| | iii. Plastic lamination AM | | | | |
| | iv. Fused Deposition Modeling | 8 | L1 | 5 | 1,3 |
| b) | Suggest suitable AM process for the following applications | | | | |
| | • Transparent Dental Aligner | | | | |
| | • Jewelry wax pattern | | | | |
| | • Color artefacts | | | | |
| | • Rocket engine | 8 | L1 | 5 | 1,3 |
| 8. a) | Explain following process with respect to surface texture improvement, accuracy improvement, aesthetic improvement | | | | |
| | • Area filling methods | | | | |
| | • Slicing & STL size | 8 | L1 | 5 | 1,3 |
| b) | List at least two engineering/medical applications each of following AM processes | | | | |
| | • Stereolithography | | | | |
| | • Ultrasonic consolidation | | | | |
| | • Selective laser melting | | | | |
| | • Wire arc additive manufacturing | 8 | L1 | 5 | 1,3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
